

SQL Code

```
1  # 1. BASIC DATA CHECKS
2  -- =====
3
4  -- Total records
5  SELECT COUNT(*) AS total_trains
6  FROM data;
7
8  -- Null checks
9  SELECT
10     SUM(`Train No`          IS NULL) AS
null_train_no,
11     SUM(`Train Name`       IS NULL OR `Train Name` = '') AS
null_train_name,
12     SUM(`Source Station`   IS NULL OR `Source Station` = '') AS
null_source,
13     SUM(`Departure Time`   IS NULL OR `Departure Time` = '') AS
null_departure,
14     SUM(`Arrival Time`     IS NULL OR `Arrival Time` = '') AS
null_arrival,
15     SUM(`Distance (km)`    IS NULL) AS
null_distance,
16     SUM(`Destination Station` IS NULL OR `Destination Station` = '') AS
null_destination,
17     SUM(`Days of Week`     IS NULL OR `Days of Week` = '') AS
null_days,
18     SUM(`Classes`         IS NULL OR `Classes` = '') AS
null_classes,
19     SUM(`Intermediate Stops` IS NULL OR `Intermediate Stops` = '') AS
null_stops
20  FROM data;
21
22  -- Duplicate trains
23  SELECT `Train No`, COUNT(*) AS cnt
24  FROM data
25  GROUP BY `Train No`
26  HAVING COUNT(*) > 1;
27  -- =====
28  -- 2. HUB ANALYSIS
29  -- =====
30
31  SELECT
32     station,
33     SUM(originating) AS originating,
34     SUM(terminating) AS terminating,
35     SUM(originating + terminating) AS hub_score
36  FROM (
37     SELECT `Source Station` AS station, COUNT(*) AS originating, 0 AS terminating
38     FROM data GROUP BY `Source Station`
39
40     UNION ALL
```

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41
42     SELECT `Destination Station`, 0, COUNT(*)
43     FROM data GROUP BY `Destination Station`
44 ) t
45 GROUP BY station
46 ORDER BY hub_score DESC
47 LIMIT 20;
48
49 -- Top routes
50 SELECT
51     `Source Station`,
52     `Destination Station`,
53     COUNT(*) AS num_trains,
54     ROUND(AVG(`Distance (km)`), 1) AS avg_distance
55 FROM data
56 GROUP BY `Source Station`, `Destination Station`
57 ORDER BY num_trains DESC
58 LIMIT 20;
59
60 -- =====
61 -- 3. ROUTE ANALYSIS
62 -- =====
63
64 SELECT
65     CASE
66         WHEN `Distance (km)` < 100 THEN 'Short'
67         WHEN `Distance (km)` < 500 THEN 'Medium'
68         WHEN `Distance (km)` < 1500 THEN 'Long'
69         ELSE 'Very Long'
70     END AS route_type,
71     COUNT(*) AS train_count,
72     ROUND(AVG(`Distance (km)`), 1) AS avg_distance
73 FROM data
74 GROUP BY route_type;
75
76 -- Longest routes
77 SELECT `Train No`, `Train Name`, `Distance (km)`
78 FROM data
79 ORDER BY `Distance (km)` DESC
80 LIMIT 10;
81
82 -- Shortest routes
83 SELECT `Train No`, `Train Name`, `Distance (km)`
84 FROM data
85 WHERE `Distance (km)` > 0
86 ORDER BY `Distance (km)` ASC
87 LIMIT 10;
88
89 -- =====
90 -- 4. SERVICE FREQUENCY
91 -- =====
92
93 SELECT
94     CASE
```

```

95         WHEN `Days of Week` LIKE '%MON%'
96         AND `Days of Week` LIKE '%TUE%'
97         AND `Days of Week` LIKE '%WED%'
98         AND `Days of Week` LIKE '%THU%'
99         AND `Days of Week` LIKE '%FRI%'
100        AND `Days of Week` LIKE '%SAT%'
101        AND `Days of Week` LIKE '%SUN%' THEN 'Daily'
102    WHEN `Days of Week` LIKE '%SAT%'
103        OR `Days of Week` LIKE '%SUN%' THEN 'Weekend Included'
104    ELSE 'Weekday Only'
105    END AS service_type,
106    COUNT(*) AS train_count
107 FROM data
108 GROUP BY service_type;
109
110 -- Day-wise trains
111 SELECT 'Monday'      AS day, COUNT(*) AS train_count FROM data WHERE `Days of Week` LIKE
'%MON%'
112 UNION ALL
113 SELECT 'Tuesday',    COUNT(*) FROM data WHERE `Days of Week` LIKE '%TUE%'
114 UNION ALL
115 SELECT 'Wednesday', COUNT(*) FROM data WHERE `Days of Week` LIKE '%WED%'
116 UNION ALL
117 SELECT 'Thursday',  COUNT(*) FROM data WHERE `Days of Week` LIKE '%THU%'
118 UNION ALL
119 SELECT 'Friday',    COUNT(*) FROM data WHERE `Days of Week` LIKE '%FRI%'
120 UNION ALL
121 SELECT 'Saturday',  COUNT(*) FROM data WHERE `Days of Week` LIKE '%SAT%'
122 UNION ALL
123 SELECT 'Sunday',    COUNT(*) FROM data WHERE `Days of Week` LIKE '%SUN%'
124 ORDER BY train_count DESC;
125
126 -- =====
127 -- 5. CLASS ANALYSIS
128 -- =====
129
130 SELECT *
131 FROM (
132     SELECT '1A' AS class_type, COUNT(*) AS total FROM data WHERE `Classes` LIKE '%1A%'
133     UNION ALL
134     SELECT '2A', COUNT(*) FROM data WHERE `Classes` LIKE '%2A%'
135     UNION ALL
136     SELECT '3A', COUNT(*) FROM data WHERE `Classes` LIKE '%3A%'
137     UNION ALL
138     SELECT 'SL', COUNT(*) FROM data WHERE `Classes` LIKE '%SL%'
139     UNION ALL
140     SELECT 'CC', COUNT(*) FROM data WHERE `Classes` LIKE '%CC%'
141     UNION ALL
142     SELECT 'GN', COUNT(*) FROM data WHERE `Classes` LIKE '%GN%'
143 ) t
144 ORDER BY total DESC;
145
146 -- Average classes per train
147 SELECT ROUND(AVG(
```

```
148         (`Classes` LIKE '%1A%') +
149         (`Classes` LIKE '%2A%') +
150         (`Classes` LIKE '%3A%') +
151         (`Classes` LIKE '%SL%') +
152         (`Classes` LIKE '%CC%') +
153         (`Classes` LIKE '%GN%')
154     ), 2) AS avg_classes
155 FROM data;
156
157 -- =====
158 -- 6. DEPARTURE TIME ANALYSIS
159 -- =====
160
161 SELECT
162     CASE
163         WHEN HOUR(STR_TO_DATE(`Departure Time`, '%H:%i')) BETWEEN 4 AND 7 THEN
164             'Early Morning'
165         WHEN HOUR(STR_TO_DATE(`Departure Time`, '%H:%i')) BETWEEN 8 AND 11 THEN
166             'Morning'
167         WHEN HOUR(STR_TO_DATE(`Departure Time`, '%H:%i')) BETWEEN 12 AND 15 THEN
168             'Afternoon'
169         WHEN HOUR(STR_TO_DATE(`Departure Time`, '%H:%i')) BETWEEN 16 AND 19 THEN
170             'Evening'
171         WHEN HOUR(STR_TO_DATE(`Departure Time`, '%H:%i')) BETWEEN 20 AND 23 THEN
172             'Night'
173         ELSE 'Late Night'
174     END AS time_slot,
175     COUNT(*) AS train_count
176 FROM data
177 GROUP BY time_slot
178 ORDER BY train_count DESC;
179
180 -- =====
181 -- 7. STOP ANALYSIS
182 -- =====
183
184 SELECT
185     `Train No`,
186     `Train Name`,
187     `Stop Count`
188 FROM data
189 ORDER BY `Stop Count` DESC
190 LIMIT 20;
191
192 -- =====
193 -- 8. FINAL SUMMARY VIEW
194 -- =====
195
196 SELECT
197     `Train No`,
198     `Train Name`,
199     `Source Station`,
200     `Destination Station`,
201     `Distance (km)`,
```

```
197      CASE
198          WHEN `Distance (km)` < 100 THEN 'Short'
199          WHEN `Distance (km)` < 500 THEN 'Medium'
200          WHEN `Distance (km)` < 1500 THEN 'Long'
201          ELSE 'Very Long'
202      END AS route_type,
203      CASE
204          WHEN `Classes` LIKE '%1A%' OR `Classes` LIKE '%2A%' THEN 'Premium'
205          WHEN `Classes` LIKE '%3A%' THEN 'AC'
206          WHEN `Classes` LIKE '%SL%' THEN 'Sleeper'
207          ELSE 'General'
208      END AS service_tier
209  FROM data;
```